

CURSOR

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MySQL 8.0 Command Line C# x + v
Query OK, 1 row affected (0.01 sec)

mysql> USE CursorDB;
Database changed
mysql> CREATE TABLE Employee (
  ->   EmpID INT PRIMARY KEY,
  ->   EmpName VARCHAR(50),
  ->   Department VARCHAR(30),
  ->   Salary INT
  -> );
Query OK, 0 rows affected (0.06 sec)

mysql> INSERT INTO Employee VALUES
  -> (101, 'Arun', 'HR', 45000),
  -> (102, 'Divya', 'Finance', 52000),
  -> (103, 'Karthik', 'IT', 60000),
  -> (104, 'Meena', 'Sales', 48000),
  -> (105, 'Rahul', 'IT', 55000);
Query OK, 5 rows affected (0.02 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> DELIMITER //
mysql> CREATE PROCEDURE DisplayEmployees()
  -> BEGIN
  ->   DECLARE done INT DEFAULT FALSE;
  ->   DECLARE id INT;
  ->   DECLARE name VARCHAR(50);
  ->   DECLARE dept VARCHAR(30);
  ->   DECLARE sal INT;
  ->   DECLARE emp_cursor CURSOR FOR
  ->     SELECT EmpID, EmpName, Department, Salary
  ->     FROM Employee;
  ->   DECLARE CONTINUE HANDLER FOR NOT FOUND
  ->     SET done = TRUE;
  ->   OPEN emp_cursor;
  ->   read_loop: LOOP
  ->     FETCH emp_cursor INTO id, name, dept, sal;
  ->     IF done THEN
  ->       LEAVE read_loop;
  ->     END IF;
  ->     SELECT id AS EmpID,
  ->            name AS EmpName,
  ->            dept AS Department,
  ->            sal AS Salary;
  ->   END LOOP;
  ->   CLOSE emp_cursor;
  -> END //
Query OK, 0 rows affected (0.06 sec)

mysql> -- END --
Query OK, 0 rows affected (0.06 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL DisplayEmployees();
+----+-----+-----+-----+
| EmpID | EmpName | Department | Salary |
+----+-----+-----+-----+
| 101 | Arun | HR | 45000 |
+----+-----+-----+-----+
1 row in set (0.01 sec)

+----+-----+-----+-----+
| EmpID | EmpName | Department | Salary |
+----+-----+-----+-----+
| 102 | Divya | Finance | 52000 |
+----+-----+-----+-----+
1 row in set (0.02 sec)

+----+-----+-----+-----+
| EmpID | EmpName | Department | Salary |
+----+-----+-----+-----+
| 103 | Karthik | IT | 60000 |
+----+-----+-----+-----+
1 row in set (0.02 sec)

+----+-----+-----+-----+
| EmpID | EmpName | Department | Salary |
+----+-----+-----+-----+
| 104 | Meena | Sales | 48000 |
+----+-----+-----+-----+
1 row in set (0.03 sec)

+----+-----+-----+-----+
| EmpID | EmpName | Department | Salary |
+----+-----+-----+-----+
| 105 | Rahul | IT | 55000 |
+----+-----+-----+-----+
1 row in set (0.03 sec)

Query OK, 0 rows affected (0.04 sec)

mysql> |
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TRIGGERS

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MySQL 8.0 Command Line Cli x + v

mysql>
mysql> CREATE DATABASE TriggerDB;
Query OK, 1 row affected (0.02 sec)

mysql> USE TriggerDB;
Database changed
mysql> CREATE TABLE Students (
  -> StudentID INT PRIMARY KEY,
  -> StudentName VARCHAR(50),
  -> Department VARCHAR(30),
  -> Marks INT
  -> );
Query OK, 0 rows affected (0.06 sec)

mysql> CREATE TABLE Student_Log (
  -> LogID INT AUTO INCREMENT PRIMARY KEY,
  -> StudentID INT,
  -> StudentName VARCHAR(50),
  -> ActionPerformed VARCHAR(50)
  -> );
Query OK, 0 rows affected (0.05 sec)

mysql> DELIMITER //
mysql> CREATE TRIGGER After_Student_Insert
  -> AFTER INSERT ON Students
  -> FOR EACH ROW
  -> BEGIN
  -> INSERT INTO Student_Log
  -> (StudentID, StudentName, ActionPerformed)
  -> VALUES
  -> (NEW.StudentID, NEW.StudentName, 'Student Record Inserted');
  -> END //
Query OK, 0 rows affected (0.03 sec)

mysql> DELIMITER ;
mysql> INSERT INTO Students VALUES
  -> (201, 'Adhavan', 'CSE', 85),
  -> (202, 'Priya', 'IT', 90),
  -> (203, 'Vijay', 'ECE', 78);
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Students;
+-----+-----+-----+-----+
| StudentID | StudentName | Department | Marks |
+-----+-----+-----+-----+
| 201 | Adhavan | CSE | 85 |
| 202 | Priya | IT | 90 |
| 203 | Vijay | ECE | 78 |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> |
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MySQL 8.0 Command Line Cli x + v

Query OK, 1 row affected (0.01 sec)

mysql> USE UpdateTriggerDB;
Database changed
mysql> CREATE TABLE Employee (
  -> EmpID INT PRIMARY KEY,
  -> EmpName VARCHAR(50),
  -> Department VARCHAR(30),
  -> Salary INT
  -> );
Query OK, 0 rows affected (0.05 sec)

mysql> INSERT INTO Employee VALUES
  -> (301, 'Kavin', 'IT', 50000),
  -> (302, 'Nisha', 'HR', 48000),
  -> (303, 'Suresh', 'Finance', 55000);
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> DELIMITER //
mysql> CREATE TRIGGER Before_Employee_Update
  -> BEFORE UPDATE ON Employee
  -> FOR EACH ROW
  -> BEGIN
  -> IF NEW.Salary < 0 THEN
  -> SET NEW.Salary = OLD.Salary;
  -> END IF;
  -> END //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> UPDATE Employee
  -> SET Salary = 50000
  -> WHERE EmpID = 301;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> SELECT * FROM Employee;
+-----+-----+-----+-----+
| EmpID | EmpName | Department | Salary |
+-----+-----+-----+-----+
| 301 | Kavin | IT | 50000 |
| 302 | Nisha | HR | 48000 |
| 303 | Suresh | Finance | 55000 |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> |
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mysql> CREATE TABLE Employee_Log (
    ->     LogID INT AUTO_INCREMENT PRIMARY KEY,
    ->     EmpID INT,
    ->     EmpName VARCHAR(50),
    ->     OldSalary INT,
    ->     NewSalary INT
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER After_Employee_Update
    -> AFTER UPDATE ON Employee
    -> FOR EACH ROW
    -> BEGIN
    ->     INSERT INTO Employee_Log
    ->     (EmpID, EmpName, OldSalary, NewSalary)
    ->     VALUES
    ->     (OLD.EmpID, OLD.EmpName, OLD.Salary, NEW.Salary);
    -> END //
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> DELIMITER ;UPDATE Employee
mysql> SET Salary = 62000

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    -> (OLD.StudentID, OLD.StudentName, OLD.Department, OLD.Marks);
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql>
mysql> SELECT * FROM Students;
+-----+-----+-----+-----+
| StudentID | StudentName | Department | Marks |
+-----+-----+-----+-----+
| 501 | Arjun | CSE | 85 |
| 502 | Lavanya | IT | 90 |
| 503 | Bharath | ECE | 78 |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql>
mysql> DELETE FROM Students
    -> WHERE StudentID = 502;
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> SELECT * FROM Students;
+-----+-----+-----+-----+
| StudentID | StudentName | Department | Marks |
+-----+-----+-----+-----+
| 501 | Arjun | CSE | 85 |
| 503 | Bharath | ECE | 78 |
+-----+-----+-----+-----+

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